

THE AUTONOMOUS INSTITUTIONAL SUBSTRATE THEOREM (AIS)

A Structural Law of Delegated Institutional Cognition, Machine-Layer Observability, and Computational Propagation

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PREFACE

The Autonomous Institutional Substrate Theorem (AIS) extends the SignalRupture (SR) Substrate Trilogy beyond continuous observation into delegated institutional cognition. Where the Continuous Reality Integration Theorem (CRI) concerns the changing temporal architecture of observation, AIS concerns the changing locus of consequential judgment.

AIS does not claim that institutions become fully autonomous, that human governance disappears, or that machine-generated error already exceeds human institutional error. Those claims cannot presently be sustained. AIS instead identifies a measurable transition already underway in bounded institutional settings: computational systems increasingly classify, prioritize, score, rank, monitor, recommend, allocate, and sometimes decide in processes whose outputs materially influence institutional behavior.

The central distinction is between tool use and cognitive delegation. A spreadsheet used by an administrator does not reorganize institutional cognition. A system that ranks cases, produces a risk score, determines queue priority, identifies fraud, recommends escalation, allocates resources, or supplies an assessment that materially shapes a legal or administrative outcome may do so. Formal human approval does not by itself establish that effective judgment remained human-centered if the computational system substantially structured the option set, classification, evidentiary framing, or priority order presented to the human decision maker.

This distinction already has an institutional analogue. The Government of Canada's Directive on Automated Decision-Making applies not only to systems that make final administrative decisions but also to systems that support or assist decisions, including assessments, recommendations, scores, summaries, and intermediate determinations presented to officers (Treasury Board of Canada Secretariat, 2025). The OECD similarly documents growing government use of AI for automation, service delivery, forecasting, resource allocation, and decision support, while identifying risks involving opacity, skewed data, accountability, and overreliance (OECD, 2025a).

AIS is therefore not a prediction that delegation will occur at some distant point. Delegation already exists. The prospective claim concerns depth: whether delegated computational functions become sufficiently consequential, coupled, and recurrent that the institution itself must be modeled as a hybrid human-machine decision system.

ABSTRACT

The Autonomous Institutional Substrate Theorem (AIS) proposes that when institutions delegate substantively consequential interpretive, classificatory, prioritization, monitoring, allocation, escalation, or decision functions to computational systems, institutional behavior increasingly emerges from coupled human-machine processes rather than from human procedure alone. AIS models machine-mediated drift, machine-layer observability, human-machine coupling, delegation depth, and computational propagation as interacting variables. It does not presume that machine-layer drift is dominant, that visibility necessarily collapses as delegation increases, or that computational harm replaces procedural harm.

Institutional drift under AIS is represented as:

$$D_AIS(t)=D_H(t)+D_M(t)+\kappa C_HM(t),$$

where D_H is human/institutional drift, D_M is machine-mediated drift, C_HM is distortion generated through human-machine coupling, and κ is coupling strength. Machine-layer observability is represented as:

$$O_M(t)=f[X(t),T(t),G(t),A(t)],$$

where X is explainability or intelligibility, T is traceability, G is grounding or validation capacity, and A is auditability. Delegation depth $A_D(t)$ measures the substantive degree to which computational outputs influence institutional outcomes.

AIS predicts that risk intensifies not because automation is inherently harmful but when consequential delegation rises while observability, grounding, accountability, or corrective capacity are insufficient. It therefore permits three outcomes: augmented institutional capacity, stable hybrid governance, and machine-mediated institutional drift.

The theorem is empirically anchored in automated public-sector decision support, algorithmic management, healthcare triage, risk scoring, automated moderation, and other systems in which computational classifications alter human or institutional action. Full AIS conditions are emerging rather than universally established.

1. INTRODUCTION — FROM TOOL USE TO DELEGATED JUDGMENT

Institutions have always relied on artifacts: forms, ledgers, categories, statistical models, databases, rules, and administrative technologies. AIS does not treat every technological aid as autonomous cognition. Its focus is narrower: systems that materially structure institutional perception or choice.

A computational system can influence governance without possessing formal authority. It may determine which cases appear urgent, which applications receive scrutiny, which transactions are flagged, which patients are classified as high risk, which workers are evaluated as underperforming, which content is escalated, or which individuals are routed into different procedural pathways. The final signature can remain human while the informational architecture of the decision has already been machine-mediated.

This is why AIS treats effective institutional judgment separately from formal authorization.

Let J_F represent formal judgment authority and J_E represent effective judgment contribution. In a purely human procedure:

$$J_E \approx J_F.$$

In a strongly delegated hybrid process:

$$J_E = J_H + J_M + J_C,$$

where J_H is direct human contribution, J_M is machine contribution, and J_C is the interaction effect produced by the structure through which machine outputs frame human choice.

AIS begins when J_M and J_C become materially consequential rather than merely clerical.

2. EMPIRICAL ANCHOR: DELEGATION IS ALREADY REAL

The empirical case for AIS is stronger than the case for a fully realized CRI society because institutions already regulate and document automated decision support.

Canada provides a direct example. The federal Directive on Automated Decision-Making defines an automated decision system broadly as technology that assists or replaces human judgment and applies to systems used to make administrative decisions or related assessments. Official guidance explicitly includes partial automation where a system provides information, assessments, recommendations, scores, summaries, or intermediate determinations that influence an officer even when the officer makes the final decision (Treasury Board of Canada Secretariat, 2025).

This is structurally important. The policy itself recognizes the AIS distinction: decision influence can be consequential before final authority transfers.

The OECD's 2025 analysis of government AI examined 200 use cases across 11 core areas of government and documented AI applications in automation, forecasting, service delivery, resource allocation, decision support, and other administrative functions. The same report emphasizes risks from skewed data, lack of transparency, overreliance, and propagated error (OECD, 2025a).

Algorithmic management provides a second anchor. A 2025 OECD employer survey covering more than 6,000 firms across France, Germany, Italy, Japan, Spain, and the United States found algorithmic management tools already widely used in most countries studied. Managers often perceived improved decision quality, while also identifying unclear accountability and difficulty following algorithmic logic as important concerns (Milanez, Lemmens, & Ruggiu, 2025).

These examples do not prove that institutions have entered a fully autonomous substrate. They demonstrate that consequential computational delegation is an observable institutional process.

3. DELEGATION DEPTH

AIS introduces Delegation Depth $A_D(t)$, a measure of how substantively a computational system participates in an institutional decision process.

Delegation should not be measured only as a binary automated/not-automated variable. A system can occupy different positions:

Level 0 — clerical assistance: formatting, retrieval, or record handling without substantive classification.

Level 1 — informational support: generates information but does not rank or recommend.

Level 2 — classificatory support: assigns categories, risk levels, scores, or anomaly flags.

Level 3 — prioritization support: ranks, triages, queues, or recommends escalation.

Level 4 — constrained decision execution: makes determinations within bounded rules subject to review.

Level 5 — autonomous decision execution: makes consequential determinations with limited contemporaneous human intervention.

These are analytical levels, not universal thresholds. They must be adapted to domain-specific law and practice.

A_D can therefore be modeled as:

$$A_D = w_1 L_c + w_2 L_p + w_3 L_a + w_4 L_r,$$

where L_c is classificatory influence, L_p is prioritization influence, L_a is action authority, and L_r is review independence. The weights w_i must be empirically calibrated.

The key AIS question becomes:

Where does effective institutional judgment occur?

not merely:

Who signs the final decision?

4. MACHINE-MEDIATED DRIFT

Machine-layer drift is an independent potential source of institutional distortion, but its dominance over human-originating drift is an empirical regime rather than a premise of AIS.

Machine-mediated drift D_M can arise when computational systems systematically diverge from the relevant institutional objective or grounded substrate through mechanisms such as proxy distortion, distribution shift, training-data inheritance, optimization mismatch, feedback loops, adversarial manipulation, unmodeled context, or repeated use outside the conditions for which the system was validated.

However, AIS does not assume:

$$D_M > D_H.$$

That empirical claim must be tested domain by domain.

The institutional drift equation is:

$$D_{AIS} = D_H + D_M + \kappa C_{HM}.$$

The coupling term matters because human and machine errors are not necessarily additive. Human operators may alter behavior in response to machine classifications. Institutions may redesign procedures around model outputs. Model developers may retrain systems on data partly generated by earlier model-mediated decisions. Human discretion may correct machine error, amplify it, or become selectively exercised around it.

Thus C_{HM} captures interactional distortion rather than assigning all failure to either humans or machines.

5. HUMAN-MACHINE COUPLING

AIS identifies human-machine coupling as a central mechanism of hybrid institutional behavior.

A simplified institutional loop is:

$$X_t \rightarrow M_t \rightarrow R_t \rightarrow A_t \rightarrow X_{t+1},$$

where X_t is the institutional environment or case population at time t , M_t is computational classification or prediction, R_t is human or procedural interpretation of that output, A_t is institutional action, and X_{t+1} is the environment modified by that action.

The system can therefore become recursive. A model does not merely describe institutional reality. Its output may alter institutional behavior, which alters the next dataset, which influences later outputs.

AIS defines coupling distortion as:

$$C_{HM}=f[R_A,O_M^{-1},F_R,P_D],$$

where R_A is reliance on algorithmic recommendations, O_M is machine-layer observability, F_R is recursive feedback intensity, and P_D is procedural dependence on machine outputs.

This formulation captures a known human-factors risk: automation bias or overreliance can cause human decision makers to overweight computational recommendations, especially when systems are treated as authoritative or when manual verification is costly. AIS does not claim this occurs universally. It identifies it as one empirically testable coupling pathway.

6. MACHINE-LAYER OBSERVABILITY

Machine-layer visibility is not uniformly lost under computational delegation; it varies with the observability of the machine layer.

AIS defines Machine-Layer Observability O_M .

$$O_M(t)=f[X(t),T(t),G(t),A(t)],$$

where:

$X(t)$ = intelligibility or explainability of the relevant output and reasoning pathway;

$T(t)$ = traceability of inputs, transformations, versions, and decision provenance;

$G(t)$ = grounding and validation against relevant real-world outcomes;

$A(t)$ = auditability by institutionally authorized reviewers or affected oversight structures.

Low O_M does not mean a system is literally invisible. It means that the institution may be unable to reconstruct why a consequential output occurred, evaluate whether it remains grounded, identify causal failure pathways, or determine whether the system continues to satisfy its intended function.

The relevant AIS risk condition is therefore not:

automation $\uparrow \Rightarrow$ visibility collapse.

It is:

$A_D\uparrow \wedge O_M\downarrow \Rightarrow$ hybrid governance risk may increase,

all else equal.

This is compatible with contemporary policy concerns. Canada's automated-decision directive emphasizes impact assessment, explainability, review, data quality, recourse, and transparency precisely because institutional use of automated systems creates new oversight demands (Treasury Board of Canada Secretariat, 2025). OECD work likewise identifies opacity and unclear accountability as recurring concerns in government and workplace algorithmic systems (OECD, 2025a; Milanez et al., 2025).

7. PROCEDURAL HARM AND COMPUTATIONAL PROPAGATION

Procedural harm remains active under computational delegation; algorithmic systems add a distinct propagation channel rather than replacing procedure.

Procedural harm remains active. Computational systems add a new propagation channel.

Let total institutional harm be:

$$H_T = H_P + H_A + H_C,$$

where H_P is procedurally generated harm, H_A is harm transmitted or replicated through automated processes, and H_C is coupling harm generated through interaction between machine outputs and institutional procedure.

AIS introduces propagation velocity:

$$v_H = \Delta N_A / \Delta t,$$

where ΔN_A is the number of consequentially affected cases, persons, or decisions over interval Δt .

The important proposition is not that every algorithmic error travels at machine speed. Many systems have human review, batching, legal constraints, or limited deployment scale. The proposition is that computational rules can replicate classifications consistently and rapidly across large case volumes, potentially altering the scale and velocity of institutional error compared with case-by-case procedural judgment.

The same architecture can also scale beneficial corrections. Once a model, rule, or validation defect is identified, a computational system may allow rapid global repair. AIS therefore treats propagation velocity as normatively neutral until coupled with the direction and severity of the propagated effect.

8. THE DELEGATION-DOMINANCE THRESHOLD

The earlier manuscript asserted that machine drift had become dominant across hybrid systems. That claim is removed.

AIS instead defines a testable machine-dominant drift ratio:

$$R_MD=[D_M+\kappa C_HM]/[D_AIS+\epsilon].$$

A machine-dominant regime can be operationally defined only after a domain establishes a justified threshold θ_MD :

$$R_MD>\theta_MD.$$

AIS specifies no universal θ_MD .

This distinction is critical. AIS does not need machine drift to dominate in order to be valid. The theorem begins when computational participation becomes structurally consequential. Dominance is a later empirical state.

9. STABLE AND UNSTABLE AIS REGIMES

AIS is not a collapse theorem. Delegation can improve institutional capacity.

A stable hybrid regime may emerge when:

- machine outputs are well grounded;
- roles are clearly defined;
- human review remains meaningful rather than ceremonial;
- model provenance and change history are traceable;
- errors can be appealed or corrected;
- performance is monitored across relevant populations;
- and automation does not suppress contextual information necessary for sound judgment.

An unstable regime may emerge when:

A_D is high;

O_M is low;

humans defer without meaningful verification;

feedback loops reinforce prior classifications;

procedures become dependent on opaque outputs;

accountability is diffused across vendors, administrators, and technical teams;

or the model's optimization target diverges from the institution's substantive obligations.

AIS therefore predicts:

AIS → augmented institutional capacity,

AIS → stable hybrid governance,

or AIS → machine-mediated institutional drift,

depending on coupling and governance conditions.

10. FORMAL THEOREM STATEMENT

Theorem — Autonomous Institutional Substrate.

When an institution delegates substantively consequential interpretive, classificatory, prioritization, monitoring, allocation, escalation, or decision functions to computational systems, institutional behavior increasingly emerges from coupled human-machine processes rather than from human procedure alone. Under these conditions, institutional drift can include human-originating, machine-mediated, and coupling-generated components; the institution acquires a distinct machine-layer observability problem; and computational systems create additional channels through which both error and correction can propagate. The magnitude and direction of resulting risk depend on delegation depth, grounding, observability, accountability, human reliance, and corrective capacity rather than on automation alone.

Formally:

$$D_{AIS} = D_H + D_M + \kappa C_{HM}$$

$$O_M = f(X, T, G, A)$$

$$R_{AIS} = f(A_D, O_M^{-1}, C_{HM}, v_H, G_v)$$

where G_v represents governance safeguards and corrective capacity.

ALS does not require $D_M > D_H$, does not require $O_M \rightarrow 0$, and does not require algorithmic harm to replace procedural harm.

11. EMPIRICAL STATUS

ALS should be classified as:

Empirically anchored emerging substrate theorem with directly observable delegation mechanisms; full hybrid-institutional dominance not established.

Supported propositions:

Institutions already deploy automated systems that make or support consequential decisions.

Computational outputs can influence human judgment even when final authority remains human.

Algorithmic management and public-sector AI are already used to automate or support tasks traditionally performed by human managers or administrators.

Opacity, accountability, explainability, data quality, and overreliance are recognized governance concerns.

Partially supported propositions:

Repeated computational delegation can reorganize institutional workflow around machine classifications.

Human-machine coupling can create recursive feedback effects not reducible to either component alone.

Computational propagation can increase the scale or velocity of repeated classifications and errors in some environments.

Prospective propositions:

Hybrid human-machine cognition becomes the dominant organizational architecture of major institutions.

Machine-mediated drift becomes larger than human-originating drift in defined sectors. Institutions become operationally dependent on autonomous agents whose outputs cannot be meaningfully reconstructed by human oversight.

12. FALSIFICATION CONDITIONS

AIS must be vulnerable to failure.

Falsifier 1 — Delegation has no structural effect.

If increases in consequential computational delegation do not alter institutional workflow, prioritization, decision distributions, review behavior, or outcome patterns, the hybridization proposition is weakened.

Falsifier 2 — Coupling is negligible.

If human and machine contributions remain statistically separable and interaction terms add no explanatory power across mature delegated systems, the C_{HM} component should be removed.

Falsifier 3 — Observability is irrelevant.

If machine-layer observability has no relationship to error detection, accountability, correction latency, or institutional trust under comparable delegation depth, O_M is not a central AIS variable.

Falsifier 4 — Computational propagation is not distinct.

If automated systems do not alter the scale, velocity, persistence, or repeatability of institutional errors compared with non-automated procedure, the propagation component must be narrowed.

Falsifier 5 — Human institutions remain effectively sovereign.

If human decision makers consistently reconstruct, override, and independently evaluate computational outputs such that machine contribution remains merely instrumental even at high automation levels, claims of a distinct AIS substrate fail in those domains.

13. GOVERNANCE CONSEQUENCES

AIS introduces a gap between formal accountability and effective decision production.

An institution can remain legally responsible while the causal pathway producing a decision becomes distributed across software, data, vendors, model developers, administrators, policies, and human reviewers. Formal authority may therefore remain centralized while effective cognition becomes distributed.

This produces several governance problems.

Accountability diffusion occurs when responsibility for an output cannot be located cleanly because multiple actors control different parts of the system.

Procedural opacity occurs when affected persons can see the final decision but cannot reconstruct the computational classification or ranking that shaped it.

Review asymmetry occurs when the nominal human reviewer lacks the time, evidence, expertise, or alternative analysis necessary to meaningfully challenge the machine output.

Delegated legitimacy risk occurs when institutions continue presenting decisions as exercises of accountable public or organizational judgment even though the substantive reasoning process has shifted into systems not equally accessible to administrators or affected populations.

AIS does not claim these failures are inevitable. It provides a diagnostic vocabulary for measuring when they appear.

14. RELATIONSHIP TO EXISTING LITERATURE

AIS overlaps with several established literatures but is not reducible to one of them.

Automation and human-factors research studies reliance, automation bias, supervisory control, and human-machine interaction.

Algorithmic governance and public-administration research studies automated decision systems, transparency, accountability, and administrative legitimacy.

Algorithmic management research studies software-mediated instruction, monitoring, evaluation, and managerial decision-making.

AI governance studies risk management, explainability, fairness, oversight, and accountability.

AIS contributes a substrate-level synthesis: it treats these phenomena as components of a common structural transition in the locus of institutional cognition. Its distinctive empirical question is not whether AI is used, but how much of effective institutional judgment has migrated into coupled computational processes.

15. RELATIONSHIP TO SR

AIS inherits SR's concern with drift, Ground-Truth Discrepancy, Slow Harm, compression, visibility, and propagation but moves them into a hybrid governance environment.

Ground-Truth Discrepancy can become machine-mediated when institutional systems repeatedly classify reality through proxies.

Silo Exposure Bias can be automated when only variables available to the model are treated as institutionally real.

Zombie Metrics can become operational inputs to automated systems, allowing outdated metrics to drive action at greater scale.

Visibility Lag can decrease at the front end through CRI-like sensing while increasing internally if the computational reasoning path becomes difficult to reconstruct.

AIS therefore creates a possible inversion:

more environmental visibility + less decision-process observability.

That inversion is one of the most important relationships in the trilogy.

16. BRIDGE FROM CRI TO AIS

CRI and AIS are analytically distinct.

CRI concerns:

Persistent Observation.

AIS concerns:

Delegated Decision.

The sequence is:

physical environment becomes more continuously observable → computational systems classify and interpret more of that environment → institutions increasingly rely on those outputs → effective judgment becomes coupled.

But one theorem does not require the other. A highly automated institution can operate on poor or episodic data. A highly instrumented environment can retain human-centered governance.

The substrate trilogy therefore describes a possible progression, not an unavoidable historical law.

17. BRIDGE TO MACS

AIS remains institution-centered. Its unit of analysis is still the hybrid institution.

The Multi-Agent Cognitive Substrate Theorem begins when that unit becomes insufficient.

If humans, institutions, models, autonomous agents, platforms, and other computational systems repeatedly observe, classify, and alter one another's outputs, then error and stability may no longer be attributable to one institution or one machine. They can become properties of the interaction network.

The bridge is therefore:

Delegated Decision → Network-Dependent Cognition.

AIS asks where institutional judgment occurs.

MACS asks what happens when judgment itself becomes distributed across recursively interacting participants.

18. CANONICAL PARAGRAPH

The Autonomous Institutional Substrate emerges when computational systems become substantively consequential participants in institutional judgment. AIS does not require full autonomy or the disappearance of human authority. It begins when systems that classify, score, prioritize, allocate, monitor, recommend, escalate, or decide materially structure institutional outcomes. Under these conditions, institutional drift becomes a hybrid property composed of human-originating, machine-mediated, and

coupling-generated components; machine-layer observability becomes a distinct governance variable; and computational systems introduce additional channels for rapid propagation of both error and correction. AIS therefore formalizes the transition from institutions that merely use technology to institutions whose effective cognition is partially produced through human-machine coupling.

19. CONCLUSION

AIS rests on three bounded propositions.

First, machine drift is not presumed dominant. Dominance is an empirical regime that must be demonstrated.

Second, machine-layer observability can rise or fall independently of delegation depth.

Third, computational harm does not replace procedural harm; it becomes an additional propagation channel interacting with existing procedure.

Together, these propositions define AIS without relying on technological determinism.

Institutions can use AI and become better. They can remain stable hybrids. They can also reorganize themselves around computational outputs in ways that create new forms of drift, opacity, reliance, and propagation.

The decisive variable is not whether a human remains formally in the loop.

It is whether the human remains substantively capable of independent judgment inside that loop.

AIS is therefore the second theorem of the SR Substrate Trilogy.

Its canonical transition is:

Delegated Decision.

The third theorem begins when no single institution, human, or autonomous system can adequately explain the behavior of the larger epistemic network in which all are participating.

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